

## C Programming

1) (a) Write an 8051 C program to find the sum of first 10 integer numbers.

```
#include <reg 51.h>
sfr P1= 0x90;
sfr A=0xE0;

void main ()
{
    unsigned char sum = 0;
    unsigned int i;
    for (i=0; i<=10;i++)
    {
        sum = sum + i;
    }
    A = sum;
    P1=sum;
}
```

(b)

```
#include <reg 51.h>
sfr P1= 0x90;
sfr A=0xE0;

void main ( )
{
    unsigned char sum = 0;
    unsigned char a[ ]= { 0x1, 0x2, 0x3,0x4, 0x5, 0x6, 0x7, 0x8, 0x9, 0xa};
    unsigned int i;
    for (i=0; i<10;i++)
    {
        sum+= a[i];

        A = sum;
        P1=sum;
    }
    while (1);
}
```

**2) Write an 8051 C program to find the factorial of a given number.**

```
#include <reg 51.h>

sfr P0= 0x80;

sfr P1= 0x90;

sfr P2= 0xA0;

sfr P3= 0xB0;

void main ( )
{
    unsigned int i;
    unsigned char num=12;
    unsigned long factorial =1;
    for (i=1; i<=num; i++)
    {
        factorial =factorial *i;
    }
    P0 =factorial;
    P1 = (factorial & 0xff00)>> 8;
    P2 = (factorial & 0xff0000)>> 16;
    P3 = (factorial & 0xff000000)>> 24;
}
```

**3) Write an 8051 C program to find square of a number (1 to 10) using Look Up Table.**

```
#include <reg51.h>

void main ( )
{
    unsigned char LUT[ ] = {1,4,9,16,25,36,49,64,81,100};
    unsigned char num, square ;
    for (num=1; num<11; num++)
    {
        square =LUT[num - 1];
        P0=square;
    }
    while (1);
}
```

**4) Write a 8051 C program to count the number of ones and zeros.**

```
#include <reg51.h>

void main( )
{
    unsigned char array [ ] = {0x57, 0xfc};
    unsigned char i,ones,zeros;
    CY=0;

    for (i=0; i<8; i++)
    {
        array[0]>>=1;
```

```
if(CY==1) ones++;  
else zeros++;  
  
}  
for(i=0; i<8; i++)  
{  
array[1]>=1;
```

```
if(CY==1)ones++;  
else zeros++ ;  
  
}  
P0=zeros;  
P1=ones;  
while(1);  
}
```